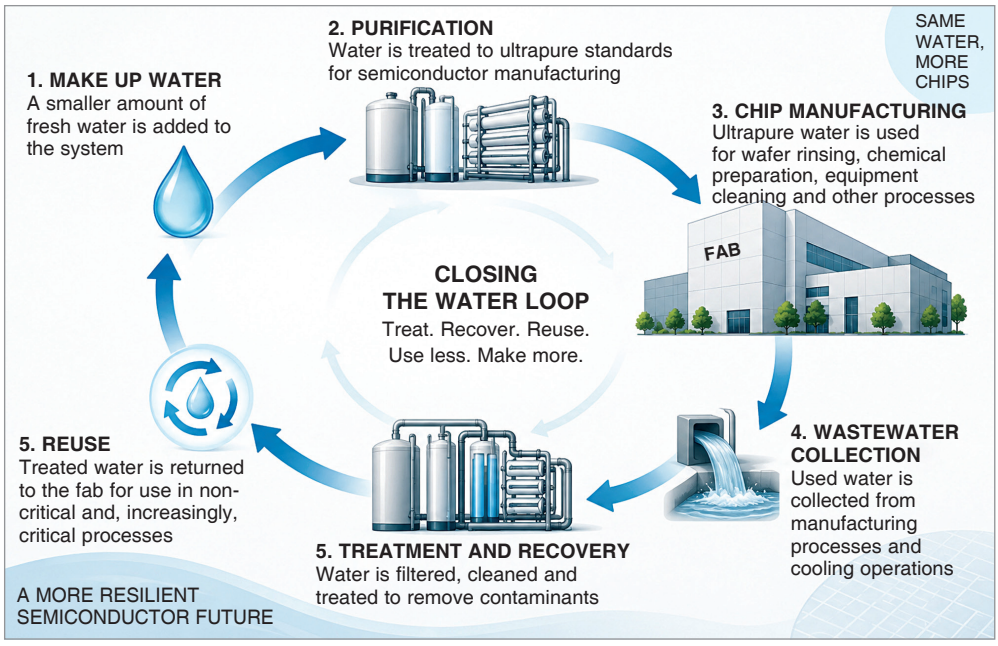


AI generated representation of a closed-loop water recycling system inside a modern semiconductor manufacturing plant

needed for manufacturing. As semiconductor production moved towards mass manufacturing, the amount of water required began to grow with it.

A 1983 US Environmental Protection Agency (EPA) assessment estimated that semiconductor manufacturing facilities in the US were using about 193 million litres of processed water every day. The water was used for wafer rinsing, chemical preparation, equipment cleaning, exhaust-gas treatment, and other manufacturing operations.

At the time, the semiconductor industry had not yet constructed today's automated mega-fabs or scaled production to billions of hyper-complex chips. Yet, the relationship between water and chipmaking had already fundamentally shifted. Water was no longer just supporting the process; it was becoming part of the process itself.

The efficiency paradox

As chips became more complex and fabs grew larger, water demand began scaling with production. More wafers, tighter manufacturing tolerances, and additional processing steps meant that the industry's water requirement was growing alongside its appetite for chips.

Intel's Arizona operations eventually required nearly 4 million gallons (15 million litres) of water a day at a single fab. The company responded by treating and reusing wastewater, recharging the aquifer, and using reclaimed municipal water for cooling and other operations. By 2008, Intel said its Arizona water programme had reduced daily water demand by up to 75 per cent.

Taiwan Semiconductor Manufacturing Company (TSMC) faced the same challenge in Taiwan, but at a much larger scale. Its water consumption reached 24.3 million cubic metres in 2011, even while it recycled 84.6 per cent of its processed water. By 2013, recycling had increased to 86.9 per cent, yet total water consumption had also risen to 29.7 million cubic metres.

That is where the industry's water problem starts to become interesting. A fab can become more water-efficient and still consume more water overall. Recycling can reduce the amount of fresh water needed for each process, but it cannot cancel out the effect of producing more wafers, building more fabs, and making increasingly complex chips.

The industry was getting better at using water, but it was also getting much better at finding reasons to use more of it. That tension would become harder to manage as semiconductor manufacturing moved into regions where water was already scarce.

A drought opens the faultlines

In 2021, Taiwan faced its worst drought in nearly seven decades, putting one of the world's biggest semiconductor manufacturing hubs under pressure. Major fabs were concentrated in Hsinchu, Taichung, and Tainan, where falling reservoir levels began to threaten industrial water supplies.

By March, Taiwan had started restricting industrial water use. Companies in two major science parks in Taichung faced a 15% reduction in supply, while restrictions were also imposed on non-industrial users in parts of Taichung and Miaoli. TSMC and Micron had fabs in the affected region, prompting manufacturers to conserve water, arrange additional tanker capacity, and secure alternative sources.

The shortage showed why a water disruption is particularly difficult for a fab to absorb. Manufacturing requires a continuous supply of extremely pure water, while TSMC's major science parks were consuming tens of thousands of tonnes every day. Despite the pressure, the company kept production running and conserved an additional 2.48 million metric tonnes of water in 2021 through wastewater recovery and other efficiency measures.

The crisis also gave more urgency to work TSMC had already